FOR AN UNRAMIFIED MORPHISM, THE RELATIVE INERTIA IS THE PULLBACK OF THE INERTIA EXACTLY WHEN AUTOMORPHISM GROUPS AGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The description of the relative inertia as sitting between the inertia of the source and the pullback of the inertia of the target. Used at the start of the counted proof (source lines 10026-10029). Produces the comparison morphism whose being an isomorphism is condition (1).
- The characterization of unramifiedness through the diagonal. Invoked inside the counted proof (source lines 10030-10032). Makes the comparison morphism etale.
- The criterion for checking a property of a morphism of algebraic stacks along a smooth cover. Invoked inside the counted proof (source lines 10033-10036) as a cross-chapter pointer. Reduces condition (1) to the comparison morphism over a presentation.
- The statement that an etale universally injective morphism of algebraic spaces is an isomorphism. Invoked at the end of the counted proof (source lines 10040-10042) as a cross-chapter pointer. Upgrades surjective universally injective etale to an isomorphism.

Source reference. Original lemma lemma-characterize-unramified: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8450–8450. Navigation pointer only: the counted text refers to the Stacks item lemma-characterize-unramified, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original remark remark-get-property-auts-from-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9992–9992. Navigation pointer only: the counted text refers to the Stacks item remark-get-property-auts-from-diagonal, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original remark remark-identify-automorphism-groups: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4046–4046. Navigation pointer only: the counted text refers to the Stacks item remark-identify-automorphism-groups, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-etale-universally-injective-open: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4656–4656. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-etale-universally-injective-open, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-check-property-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 479–479. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-check-property-covering, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be an unramified morphism of algebraic stacks. The following are equivalent*

(1) *$\mathcal{I}_\mathcal{X} \to \mathcal{X} \times_\mathcal{Y} \mathcal{I}_\mathcal{Y}$ is an isomorphism, and*
(2) *f induces an isomorphism between automorphism groups at x and $f(x)$ (Remark remark-identify-automorphism-groups) for all $x \in |\mathcal{X}|$.*

Proof. Choose a scheme U and a surjective smooth morphism $U \to \mathcal{X}$. Denote $G \to H$ the pullback of the morphism $\mathcal{I}_\mathcal{X} \to \mathcal{X} \times_\mathcal{Y} \mathcal{I}_\mathcal{Y}$ to U. By Remark remark-get-property-auts-from-diagonal and Lemma lemma-characterize-unramified the morphism $G \to H$ is étale. Condition (1) is equivalent to the condition that $G \to H$ is an isomorphism (this follows for example by applying Properties of Stacks, Lemma stacks-properties-lemma-check-property-covering). Condition (2) is equivalent to the condition that for every $u \in U$ the morphism $G_u \to H_u$ of fibres is an isomorphism. Thus (1) $\Rightarrow$ (2) is trivial. If (2) holds, then $G \to H$ is a surjective, universally injective, étale morphism of algebraic spaces. Such a morphism is an isomorphism by Morphisms of Spaces, Lemma spaces-morphisms-lemma-etale-universally-injective-open. □